Forest & Wildlife

Problem Statement :

Automated Camera Trap Triage and Individual Tiger Movement Intelligence System for Pench Tiger Reserve

Background

Pench Tiger Reserve deploys camera traps across hundreds of grid-based stations during each monitoring cycle. A single cycle generates tens of thousands of images, of which a large majority are false triggers — moving grass, heat shimmer, insects, rain, or shifting light. Field staff and researchers currently sort these images manually, spending hundreds of person-hours simply to discard blanks before any meaningful analysis begins. The work that follows is equally manual. Identifying which individual tiger appears in a photograph requires visually comparing flank stripe patterns against a reference catalogue, a slow and error-prone process. Determining where each tiger ranges, and noticing when an individual's movement pattern changes, currently depends on institutional memory rather than systematic analysis. By the time a shift is recognised — a tigress moving into new territory, a sub-adult dispersing toward buffer villages, or an animal simply ceasing to appear — the window for a management response has often passed.

Objective

Participants must develop software that accepts raw, unprocessed camera trap image folders exactly as they come off the field SD cards, and produces a clean individual-tiger database along with area occupancy and movement-deviation intelligence — with minimal human intervention.

Required Functional Outputs

1. Blank image filtering and removal

The system must ingest raw image directories and automatically classify each frame as blank or containing a subject. Blank images must be removed from the working dataset. Deletion must be safe and reversible — a quarantine or staged-delete approach with a confidence threshold is expected, since irreversible deletion of a misclassified frame destroys irreplaceable field data. The system must report how many frames were removed and the space and time saved.

2. Individual tiger identification via stripe pattern, building a persistent database

For every retained image containing a tiger, the system must detect the animal, isolate the flank, and extract the stripe pattern to match against a growing catalogue of known individuals. New individuals must be automatically enrolled with a new identifier. Confident matches should be applied automatically; ambiguous matches must be surfaced to a human reviewer rather than silently guessed. The result is a persistent, queryable database linking each identified individual to every image, station, timestamp, and GPS location where it was captured.

3. Tiger-wise area occupancy, regenerated on every run

After each processing run, the system must generate for every individual: the set of locations where it was captured, a mapped home range estimate, the centroid of its activity, and the estimated area occupied. This must be visualised on a tiger reserve map and exported in a form usable by forest department staff. Overlap between individuals must be visible, since territorial overlap is itself a management signal.

4. Deviation and trend alerting

The system must compare each run against the individual's established history and raise alerts for meaningful change. At minimum this should cover: a shift of range centroid beyond a defined threshold (15-20 sq km in core and 5 kms in the buffer region), first capture at a station the individual has never used, movement into or toward buffer or village-adjacent stations, and prolonged absence of a previously regular individual. Alerts must distinguish genuine behavioural deviation from artefacts of uneven survey effort — a tiger appearing at a new station because a new camera was installed there is not a movement deviation. Each alert must state what changed, the supporting evidence, and a confidence level.

Constraints and Design Requirements

The software must run on ordinary field hardware — a standard laptop without a dedicated GPU and without internet connectivity, since processing frequently happens at a forest rest house or range forest office. Batch processing of tens of thousands of images must complete in a practical timeframe. Camera clock drift, reset timestamps, inconsistent folder naming, and mixed-up SD cards are normal field realities and the system must handle or flag them rather than fail. Images capturing humans must be handled with appropriate privacy safeguards. The interface must be usable by forest department staff who are not data scientists, and every automated decision must be auditable and correctable by a human.

Evaluation Criteria

Solutions will be assessed on blank-detection accuracy with particular attention to false negatives, individual identification accuracy against a held-out reference set, quality and interpretability of the occupancy output, whether alerts are genuinely actionable rather than noisy, processing throughput on constrained hardware, robustness against messy real-world input, and overall usability by the intended end user.

Expected Deliverables

A working end-to-end pipeline demonstrated on a sample raw image set, the individual database with its schema, map-based occupancy visualisation, a functioning alert output, and documentation covering setup, model choices, and known limitations.